Incident report INC-2026-007: oscillation of the product separator pressure PIC-1002

Unit: U-200 Hydrotreating · Equipment: V-1004 Product separator · Date of event: 2026-02-03 · Report issued: 2026-02-12 · Classification: process upset, no release, no injury

Summary

On 2026-02-03, between 02:10 and 03:40, the product separator pressure PIC-1002 oscillated with a growing amplitude, from ±1 bar to ±4 bar, with a period of about 3 minutes. The oscillation passed to the separator level LIC-1004, which reached its high alarm (95 %) twice, and to the surge drum V-1105. The shift put PIC-1002 in manual at 03:40 and the pressure settled in 10 minutes.

Sequence of events

- 01:55 The charge was raised from 60 to 64 m^3/h on FIC-1201.
- 02:10 PIC-1002 started to oscillate. The amplitude grew over the next 30 minutes.
- 02:45 LIC-1004 high alarm (95 %). The outlet valve LV-1004 was already at 100 % output.
- 03:20 Second LIC-1004 high alarm.
- 03:40 PIC-1002 put in manual by the panel operator at 48 % output.
- 03:50 Pressure stable at 52 bar.

Causes

1. Direct cause. The controller gain of PIC-1002 was raised from 1.6 to 2.4 in the 2025 turnaround (tuning revision B of DOC-CN-001) to speed the recovery after upsets. With the dead time of about 28 s of this loop, the new gain leaves too little stability margin. A charge increase is enough to start a sustained oscillation.
2. Contributing cause. The separator level valve LV-1004 is at its upper limit at high throughput (saturation, ER-25-044), so the level cannot absorb the pressure swings.
3. A valve travel test on PV-1002 on 2026-02-09 showed a dead band of 0.3 %, so valve stiction was ruled out.

Actions

Action	Owner	Due	Status
Reduce the controller gain of PIC-1002 by 30 % and verify with a step test	Process control engineering	2026-03-31	open
Larger trim for LV-1004 (ER-25-044)	Process engineering	next turnaround	open
Add the oscillation case to the operating	Operations	2026-03-15	done

Action	Owner	Due	Status
manual (DOC-U200-OM-001, section 9.2)			

Lessons

A tuning change made to speed one response must be checked against the stability margin at all throughputs, not only at the throughput of the test.